

Finite-Response Structure in Language-Model Grokking

Apparent takeoff as a resolved, two-stage generalization process

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Abstract

Resource scaling laws vary model size, data, or compute. We study a different object: the optimization-time path of delayed generalization in a small causal Transformer. The cumulative arrival of improvement is a step response F_t , and its finite difference κ_t is a response kernel. A delta kernel is instantaneous; a broad or multilobed kernel carries temporal structure that a terminal exponent or changepoint cannot retain. Across six independent seeds on modular addition, the late rise from 50 to 90 percent held-out accuracy occupies 600–1,250 optimizer updates, or 12–25 resolved checkpoint intervals. More strikingly, a basis-free valley theorem rejects every single unimodal response kernel on checkpoints withheld from selecting the valley, in all six seeds. A two-response mixture then predicts those held-out checkpoints better than either a step or one smooth response in every seed. Roughly 30 percent of the fitted response arrives in an early component and 70 percent in a later, sharper component. Apparent takeoff is therefore the second lobe of a longer response. A separate modal test finds no need for two exponential settling modes, so multiplicity of the transition does not yet identify two physical mechanisms.

1. What a terminal curve leaves out

Neural scaling laws compress families of models, datasets, or compute budgets into predictive resource–performance relations [1, 2]. Smoothly broken power laws extend that program to inflections and delayed transitions across scale [4]. Our independent variable is not resource scale: architecture and data fraction are fixed within each run, and t is optimizer update. We therefore study a *training trajectory*, using scaling laws only as a motivating contrast. A terminal rate still does not say whether improvement turns on immediately, gathers momentum, or arrives through overlapping stages.

This missing temporal information matters most when a curve is described as *takeoff*. A late transition can look sudden simply because its waiting time is long compared with its final rise. That contrast is visually real, but it does not make the rise instantaneous. Nor does the location of a changepoint tell us whether earlier, smaller components were already delivering part of the eventual gain. Metric choice can itself create apparent emergence [5]; any temporal claim must therefore name both its score and its observation resolution.

The finite-response view separates three questions:

1. What terminal law, if any, does the capability coordinate approach?
2. Over what interval does the new response arrive?
3. Is that response one kernel, a mixture of parallel kernels, a convolution of sequential stages, or a sum of settling modes?

The first is a resource-scaling question. The second and third concern the training path and are the subject of this paper.

The empirical result below gives the distinction a concrete form. In a small causal Transformer, held-out generalization does not wait motionless and then jump. It first rises to a partial plateau, continues through a long shallow middle, and then completes in a comparatively sharp second rise. That pattern survives six new seeds and a nonparametric test on checkpoints excluded from selecting the pattern. The scientifically interesting object is therefore not a magic iteration. It is the shape by which learnable structure becomes visible.

2. From trajectory to takeoff kernel

For a positive, unbounded capability coordinate C_t , define logarithmic velocity

$$u_t = \log C_{t+1} - \log C_t. \quad (1)$$

If $u_t \rightarrow \alpha > 0$, normalize the velocity and difference it:

$$F_t = \frac{u_t}{\alpha}, \quad \kappa_t = F_t - F_{t-1}, \quad F_{-1} = 0. \quad (2)$$

Then F is the observed step response and κ is its causal takeoff kernel. An ideal jump at time τ has

$$F_t = H_{t-\tau}, \quad \kappa_t = \delta_{t\tau}. \quad (3)$$

Any mass spread across more than one resolved interval is a finite response, not a delta.

Accuracy on a fixed benchmark is bounded and cannot sustain positive exponential velocity. For a bounded score J_t with endpoints $J_0 < J_\infty$, the appropriate object is instead

$$F_t^b = \frac{J_t - J_0}{J_\infty - J_0}, \quad \kappa_t^b = F_t^b - F_{t-1}^b. \quad (4)$$

This kernel records when the eventual score gain is delivered. It does not claim that accuracy itself obeys an asymptotic exponential law. The experiment uses (4); equations (1)–(2) remain the natural version for an unbounded coordinate such as inverse error or a task frontier. Following the metric sensitivity emphasized by Schaeffer et al. [5], all claims below are explicitly conditional on held-out accuracy and 50-update sampling. The same need to resolve local timing now appears at practical scale: one-pass mixture-of-experts pretraining has shown asynchronous memorization-to-generalization transitions across data groups [12].

Proposition 1 (Normalization and resolution). *If $F_t \rightarrow 1$, then $\sum_{t \geq 0} \kappa_t = 1$ in convergent partial sums. If F_t is nondecreasing, κ is a probability distribution over response time. At checkpoint spacing Δ , however, a delta and any response confined inside one checkpoint interval are observationally indistinguishable from trajectory samples alone.*

Proof. Telescoping gives

$$\sum_{t=0}^T \kappa_t = \sum_{t=0}^T (F_t - F_{t-1}) = F_T.$$

Consequently,

$$\lim_{T \rightarrow \infty} \sum_{t=0}^T \kappa_t = \lim_{T \rightarrow \infty} F_T = 1.$$

If the response profile is nondecreasing, then each kernel entry satisfies

$$\kappa_t = F_t - F_{t-1} \geq 0.$$

Finally, observations at spacing Δ reveal only the mass delivered in each sampled interval,

$$F_{(j+1)\Delta} - F_{j\Delta} = \sum_{j\Delta < t \leq (j+1)\Delta} \kappa_t.$$

A delta and any kernel confined to that interval with the same total mass therefore induce the same sampled response. $\square$

For a nonnegative kernel, the center, variance, and effective support are

$$\mu_\kappa = \sum_t t \kappa_t, \quad \sigma_\kappa^2 = \sum_t (t - \mu_\kappa)^2 \kappa_t, \quad W_{\text{eff}} = \exp \left[- \sum_t \kappa_t \log \kappa_t \right]. \quad (5)$$

A delta has zero variance and $W_{\text{eff}} = 1$. These statistics turn *sudden* into a resolution-dependent measurement rather than a visual impression.

Theorem 1 (Terminal laws do not identify takeoff). *For every $L \geq 1$, there exist normalized profiles with the same terminal velocity α such that one has a delta takeoff kernel and another has a kernel supported on L distinct rounds.*

Proof. For the immediate profile, take

$$\kappa_t^{(0)} = \delta_{t0}.$$

Choose a positive unit-mass sequence

$$q_t > 0 \quad (0 \leq t < L), \quad \sum_{t=0}^{L-1} q_t = 1,$$

and define the finite-width kernel by

$$\kappa_t^{(L)} = \begin{cases} q_t, & 0 \leq t < L, \\ 0, & t \geq L. \end{cases}$$

For either construction $a \in \{0, L\}$, form its cumulative response

$$F_T^{(a)} = \sum_{t=0}^T \kappa_t^{(a)}, \quad \lim_{T \rightarrow \infty} F_T^{(a)} = 1.$$

Now set

$$u_t^{(a)} = \alpha F_t^{(a)}.$$

Both profiles have the same terminal velocity,

$$\lim_{t \rightarrow \infty} u_t^{(0)} = \lim_{t \rightarrow \infty} u_t^{(L)} = \alpha,$$

while their kernels have support sizes one and L , respectively. $\square$

The theorem is elementary because the identifiability failure is elementary. An exponent is terminal information. A kernel is path information.

3. One response, two responses, or two mechanisms?

Before choosing a parametric response, one mode has a basis-free obstruction. Call a nonnegative kernel *unimodal* if it is nondecreasing up to some mode and nonincreasing thereafter.

Theorem 2 (The single-mode valley obstruction). *Let $I_1 < I_2 < I_3$ be three ordered, disjoint intervals and let*

$$\bar{\kappa}(I) = \frac{1}{|I|} \int_I \kappa(t) dt.$$

If

$$\bar{\kappa}(I_2) < \min\{\bar{\kappa}(I_1), \bar{\kappa}(I_3)\},$$

then the nonnegative kernel κ is not unimodal. The same statement holds for a discrete kernel with interval integrals replaced by sums.

Proof. The interval averages imply the existence of ordered points $t_1 < t_2 < t_3$ for which

$$\kappa(t_1) \geq \bar{\kappa}(I_1), \quad \kappa(t_2) \leq \bar{\kappa}(I_2), \quad \kappa(t_3) \geq \bar{\kappa}(I_3).$$

Hence

$$\kappa(t_2) < \min\{\kappa(t_1), \kappa(t_3)\}.$$

But a unimodal kernel obeys

$$\kappa(t_2) \geq \min\{\kappa(t_1), \kappa(t_3)\}$$

for every ordered triple: if its mode lies before t_2 , the right side is bounded by $\kappa(t_3)$; if after t_2 , it is bounded by $\kappa(t_1)$. This is a contradiction. $\square$

For equal-width intervals of width w , define the valley certificate

$$V_w = \min\{\bar{\kappa}(I_1), \bar{\kappa}(I_3)\} - \bar{\kappa}(I_2).$$

The certificate is deterministically robust. If an estimated cumulative profile satisfies $|\hat{F} - F| \leq \varepsilon$ at the six window endpoints, then each average rate changes by at most $2\varepsilon/w$. Consequently,

$$\hat{V}_w > \frac{4\varepsilon}{w}$$

still rules out one mode. This result establishes multiplicity without deciding whether the cause is a mixture, a sequence, or something else.

Finite responses combine in two different ways. Parallel channels mix:

$$\kappa_{\text{mix}}(t) = \sum_{r=1}^m \pi_r \kappa_r(t), \quad \pi_r \geq 0, \quad \sum_r \pi_r = 1. \quad (6)$$

Sequential independent stages convolve:

$$\kappa_{\text{seq}} = \kappa_m * \cdots * \kappa_2 * \kappa_1. \quad (7)$$

Addition means that several routes deliver portions of the response. Convolution means that one response must pass through successive delays. The algebra is not cosmetic: mixtures can produce separated lobes, whereas convolution adds stage delays and spread.

A third decomposition concerns local settling modes. After an onset τ , one may fit

$$Y_t = F_t - 1 = \sum_{r=1}^m c_r \theta_r^{t-\tau}, \quad |\theta_r| < 1. \quad (8)$$

Real positive modes settle monotonically; negative or complex modes alternate or ring. In noiseless data the minimal mode count is the rank of a sufficiently large Hankel matrix, and the modes are recoverable by the usual Prony or matrix-pencil construction. But modes are properties of the curve, not automatic labels for hidden mechanisms. A single nonlinear mechanism may generate several modes, and several mechanisms may collapse to one.

This distinction sets the evidentiary bar. A second component should improve prediction outside the checkpoints used to fit it, survive independent seeds, and remain resolved above checkpoint spacing. Even then, it is a response stage until an intervention gives it a mechanistic name.

4. A dense, inexpensive test

We trained a decoder-only causal language model from random initialization on modular addition, following the small algorithmic setting in which grokking was first isolated [3]. Each input was the token sequence $[a, +, b, =]$; the next-token target was $(a + b) \bmod 67$. The model had one Transformer block, width 64, four attention heads, feed-forward width 256, and 59,136 trainable parameters.

This setting now has several complementary explanations. Mechanistic progress measures resolve memorization, circuit formation, and cleanup [6]; theory links delayed generalization in modular addition to leaving an early kernel-like regime [7]; and broader algorithmic experiments connect grokking to competition between memorizing and generalizing circuits [8]. Our analysis is deliberately output-side: it asks what temporal structure the held-out trajectory identifies before assigning a circuit-level cause. More generally, delayed generalization has been attributed to a transition from lazy to rich feature learning [9] and, in homogeneous networks, to distinct early- and late-phase implicit biases [10].

All 4,489 ordered pairs were available. A four-condition pilot varied training fraction and weight decay, then selected the condition with the longest completed generalization delay: 30 percent training data (1,347 examples), 70 percent held out (3,142 examples), and AdamW weight decay one. Six fresh seeds

then independently changed initialization and the data split. This pilot choice makes the confirmation conditional on a deliberately selected delayed-generalization regime; it is not an unbiased prevalence estimate. The optimizer and regularization choice is substantive: numerical stability and gradients that scale logits without changing predictions can alter or prevent grokking [11].

Training used learning rate 10^{-3} , batch size 512, and full train/test evaluation every 50 optimizer updates. For every seed we compared:

$$\begin{aligned} \text{step:} & \quad p_0 + (p_1 - p_0)H(t - \tau), \\ \text{one response:} & \quad p_0 + (p_1 - p_0)S((t - \tau)/s), \\ \text{two responses:} & \quad p_0 + (p_1 - p_0)[\pi S((t - \tau_1)/s_1) + (1 - \pi)S((t - \tau_2)/s_2)], \end{aligned} \tag{9}$$

where $S(x) = (1 + e^{-x})^{-1}$. Correct counts supplied binomial weights for fitting. Because the same held-out examples recur at every checkpoint, checkpoint observations are temporally dependent; we therefore selected models with checkpoint-level

$$\text{BIC}_{\text{cp}} = n \log(\text{RSS}/n) + k \log n \tag{10}$$

and RMSE on alternating checkpoints excluded from fitting, rather than treating every repeated test answer as an independent longitudinal sample. Synthetic delta, single-response, and two-response trajectories were recovered before analyzing the experiment.

We also tested one mode without fitting a response family. On even-numbered checkpoints, three ordered, nonoverlapping windows were chosen to maximize V_w . Their locations were then frozen and V_w was evaluated on the odd-numbered checkpoints after a separate weighted isotonic projection. The primary window width was 1,000 optimizer updates; widths 800 and 1,500 tested scale sensitivity. A noiseless logistic and a separated two-response curve verified that the procedure respectively withholds and produces a positive certificate.

The complete cloud workload took 392 seconds on one CPU instance. Including a failed launcher attempt, compute, temporary storage, and collection cost about \$0.10. Raw trajectories, scripts, hashes, and post-processing provenance are retained with the manuscript.

5. The jump is resolved

The first empirical fact is delay. Training accuracy reached 99 percent at update 350 or 400 in every seed. Held-out accuracy did not reach 50 percent until updates 3,900–13,450. The system had already fit its observed examples; what followed was the delayed appearance of a rule on unseen pairs.

The second fact is width. From the first observed 50 percent accuracy to the first observed 90 percent took 600–1,250 updates, with median 1,000. At 50-update spacing, the visibly steep final rise alone occupies 12–25 resolved intervals. The full isotonic bounded-response kernel is broader still: its 10–90 percent mass width is 4,262–9,611 updates, with median 5,781 and a median effective support of 23.5 checkpoints.

Quantity	Minimum	Median	Maximum
Training 99% step	350	400	400
Held-out 50% step	3,900	5,025	13,450
Observed 50–90% width	600	1,000	1,250
Full-kernel 10–90% width	4,262	5,781	9,611
Effective kernel checkpoints	19.4	23.5	29.9
Held-out valley V_{1000}	0.049	0.134	0.215
Uniform-error radius	1.2 pp	3.3 pp	5.4 pp

Table 1: Resolved timing over six confirmation seeds. Widths are optimizer updates; the observation spacing is 50 updates. The valley is normalized response gain per 1,000 updates; the final row is the endpoint perturbation required to erase its sign.

Thus *sudden* is true only comparatively: the last rise is short beside the preceding delay. It is not

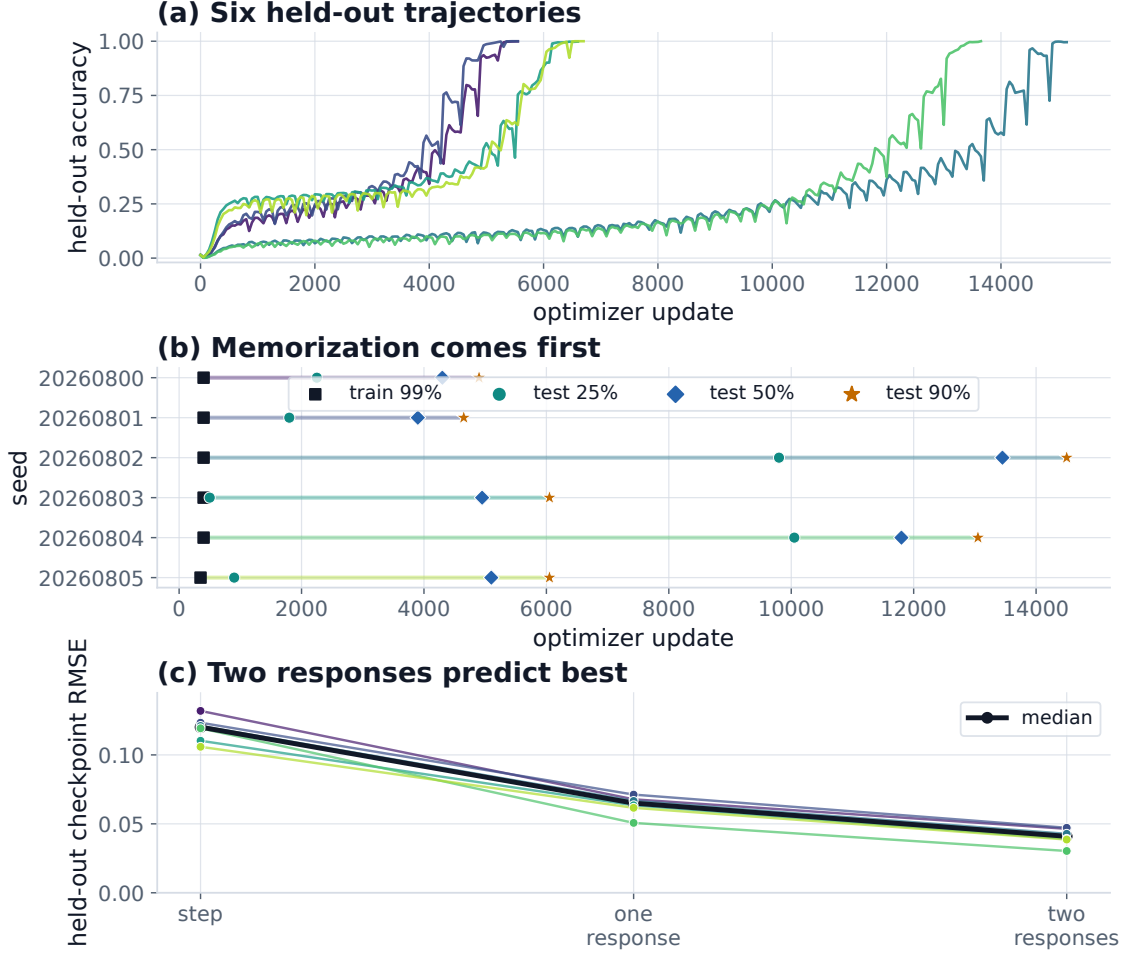


Figure 1: **The overall experiment.** (a) Held-out accuracy for six confirmation seeds. (b) Training accuracy reaches 99 percent hundreds to thousands of updates before held-out thresholds; the 50–90 percent segment itself spans many checkpoints. (c) Checkpoint-level cross-validated error falls for every seed from a step, to one smooth response, to two responses.

instantaneous at the available resolution. The takeoff impression comes from juxtaposing two timescales, not from a delta.

6. One mode fails; two responses explain why

Width alone does not explain the shape in Fig. 1(a). Each trajectory contains an early partial rise and a later completion. The nonparametric test makes that observation exact: the held-out V_{1000} is positive in all six seeds, ranging from 0.049 to 0.215 normalized response per 1,000 updates. All 18 seed-by-width evaluations remain positive at widths 800, 1,000, and 1,500. Thus the observed high–low–high delivery pattern cannot arise from any single nonnegative unimodal kernel at these resolutions. The weakest certificate survives arbitrary endpoint errors up to 1.2 percentage points; the median survives 3.3 points.

Having established multiplicity without a basis, we use the two-response fit to describe it economically. Write

$$F_i(t) = S\left(\frac{t - \tau_i}{s_i}\right), \quad \kappa_i(t) = \frac{dF_i}{dt}. \quad (11)$$

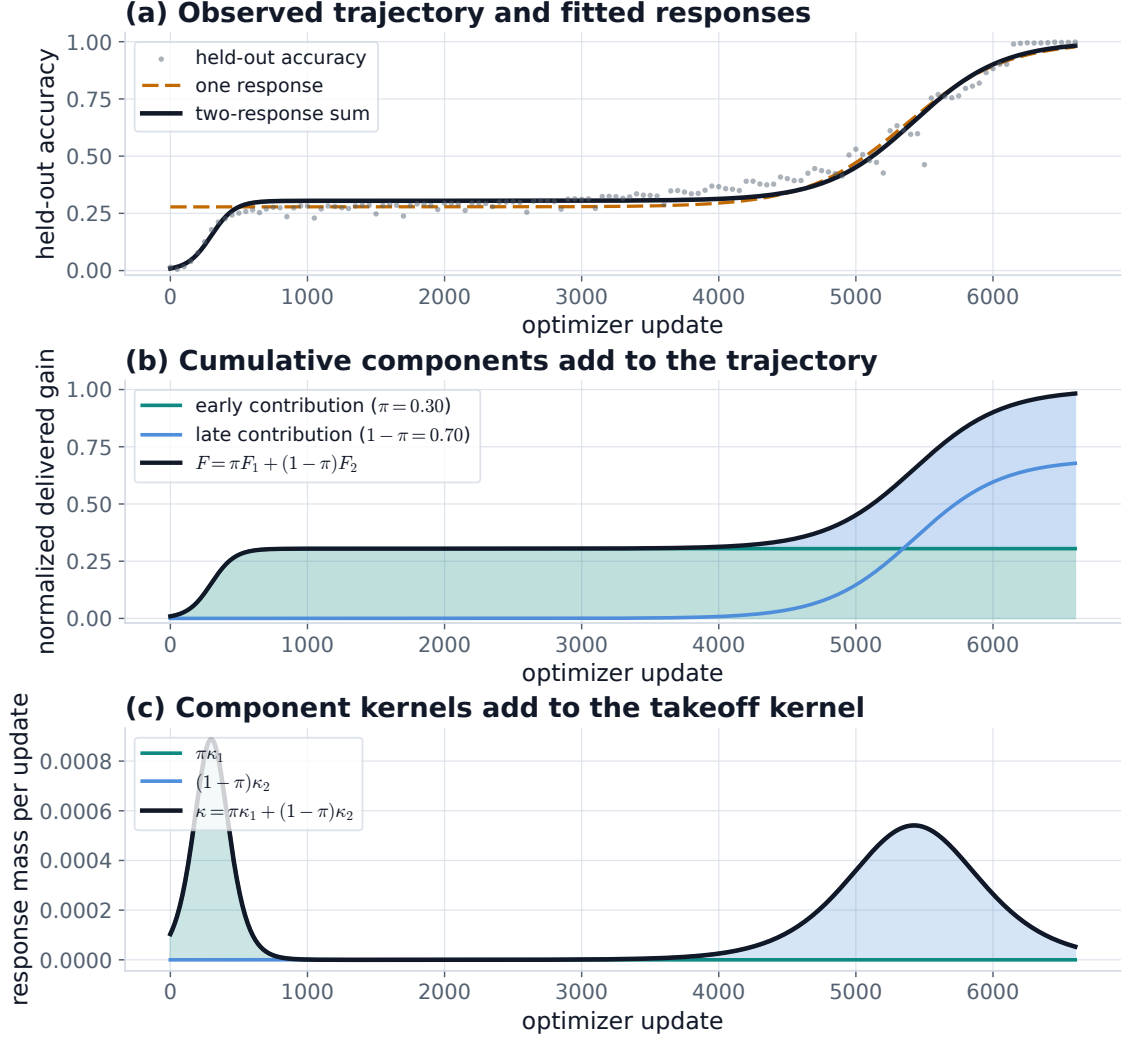


Figure 2: **Anatomy of the fitted response.** (a) For representative seed 20260803, one response absorbs the early plateau into a baseline; two responses follow the data. (b) The early and late cumulative contributions add to the normalized trajectory. (c) Their weighted kernels add pointwise to the final takeoff kernel. This is the parallel mixture (6), not a demonstrated sequential convolution (7).

Then the fitted trajectory and its derivative obey the same mixture law:

$$\hat{F}(t) = \pi F_1(t) + (1 - \pi) F_2(t), \quad \hat{\kappa}(t) = \pi \kappa_1(t) + (1 - \pi) \kappa_2(t). \quad (12)$$

Within this descriptive model, the final trajectory is literally the cumulative sum of two weighted finite-response kernels. Integrating the right side of (12) gives the two-stage curve. Figure 2 displays that identity directly: the cumulative components add in panel (b), and their differentiated kernels add in panel (c).

For the representative seed, the early component is centered at update 298 and carries 30.5 percent of the response; the late component is centered at 5,424 and carries 69.5 percent. Across all seeds, early mass ranges from 26.5 to 40.4 percent, with median 30.2 percent. The component centers are separated by 3,128–5,126 updates (median 3,824). The early component varies from sharp to highly diffuse, while the late component is strikingly more stable: its fitted 10–90 percent width lies between 1,193 and 1,441 updates in every seed.

Trajectory model	Parameters	Held-out checkpoint RMSE	BIC wins
Delta step	3	0.106–0.132	0/6
One logistic response	4	0.051–0.071	0/6
Two-response mixture	7	0.030–0.047	6/6

Table 2: Predictive comparison. RMSE is evaluated on alternating checkpoints excluded from fitting; BIC treats checkpoints, not repeated evaluation items, as longitudinal observations.

The theorem and the fit answer complementary questions. The held-out valley rules out one mode without selecting a basis; the mixture is favored by both checkpoint BIC and held-out checkpoint RMSE in all six seeds and supplies a compact decomposition. Yet a different question receives a negative answer. We fit one and two decaying exponentials to the post-onset residual over both the full tail and a local 5–95 percent window. The penalized fit selects one mode in all 12 comparisons. The data support two separated *arrivals* of generalization, not two exponential tail modes. That negative result is useful: it locates the extra structure in the response mixture rather than multiplying hidden mechanisms without evidence.

7. What the experiment changes

The strongest claim is narrow enough to be defensible and broad enough to matter:

In this controlled language model, apparent takeoff is the terminal lobe of a temporally extended generalization response. It is neither instantaneous nor compatible with any single nonnegative unimodal kernel at the tested resolutions; a two-response mixture predicts and describes its two stages.

The claim is novel within this framework because it joins two facts that are usually reported separately. Grokking establishes a delay between memorization and generalization. Finite-response analysis resolves what happens *inside* that delayed transition. A held-out valley first establishes that the route is not one mode; the fitted decomposition then shows that about one-third of the eventual held-out gain arrives through an earlier component, while a later, comparatively stable kernel delivers the rest. A single *grokking time* marks the most visible lobe and misses the response already in motion.

This suggests a sharper reading of language-model training trajectories that appear emergent. A changepoint is temporally incomplete; a kernel adds center, width, entropy, and internal composition. It also forces the observation scale into the claim: a response may be delta-like at coarse resolution and plainly finite at dense resolution.

The experiment does not establish a universal language-model kernel. It uses one small architecture, one synthetic task, one pilot-selected hyperparameter condition, and six seeds. Alternating checkpoint validation reduces but does not remove temporal dependence. The logistic components are an economical descriptive basis, not discovered neurons or circuits. Most importantly, a mixture fit does not prove two physical channels. Distinguishing mixture from convolution, or assigning components to memorization, representation formation, and rule deployment, requires interventions and intermediate-state measurements. The valley certificate is likewise conditional on a nonnegative latent response, the isotonic projection, and the stated window scale; it proves a shape constraint, not a mechanism.

The next decisive test is therefore comparative rather than merely larger: predeclare the coordinate and estimators, then vary task family, data fraction, regularization, depth, width, and model scale. If centered kernels collapse, the common shape becomes a candidate universality class. If the approximately 30/70 split or the stable late width persists, it becomes a law worth explaining. If not, the kernel still does its job: it shows exactly how the route differs when the endpoint does not.

8. Conclusion

An asymptote says where a learning curve goes; a finite-response kernel says how it arrives. Here the late jump is real but neither singular nor alone. A held-out certificate rules out one unimodal response,

and a predictive two-response model shows how the gain separates: generalization begins earlier, travels through a broad first component, and completes through a sharper second. The scientific object is not an instant of takeoff, but the structure hidden by calling the journey a jump.

Conceptual takeaways

- A long wait can make a gradual transition feel sudden. Zoom in on the training timeline, and the apparent jump becomes a response with measurable width and shape.
- In these runs, generalization was already under way before the dramatic late rise: an early component delivered roughly one-third of the eventual gain, and a sharper later component delivered the rest.
- A single “takeoff time” names the most visible moment but misses the route. A response kernel keeps track of when improvement arrives, how spread out it is, and whether it comes in one wave or several.
- Two waves in the learning curve do not, by themselves, prove two mechanisms inside the model. Establishing that link requires interventions and measurements of the model’s intermediate states.

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